```
clc;
clear;
close all;

%% PARAMETERS
A = 1;
f = 50;
Fs = 1000;
N = 1000;

n = 0:N-1;
x = A*sin(2*pi*f*n/Fs);

B = 8;
bits = [4 8 12 16];

%% 1. 8-BIT QUANTIZATION
D = 2*A/2^B;

xr = D*round(x/D);     % Rounding
xt = D*fix(x/D);       % Truncation

er = x - xr;           % Rounding error
et = x - xt;           % Truncation error

SQr = 10*log10(mean(x.^2)/mean(er.^2));
SQt = 10*log10(mean(x.^2)/mean(et.^2));

%% 2. INPUT / OUTPUT QUANTIZATION
xq = xr;
y = x;
yq = D*round(y/D);

eo = y - yq;
SQout = 10*log10(mean(y.^2)/mean(eo.^2));

%% 3. WORD-LENGTH COMPARISON
step = zeros(size(bits));
sqnrR = zeros(size(bits));
sqnrT = zeros(size(bits));
maxER = zeros(size(bits));
maxET = zeros(size(bits));

for k = 1:length(bits)
    b = bits(k);
    d = 2*A/2^b;

    qr = d*round(x/d);
    qt = d*fix(x/d);
```

```
    er1 = x - qr;
    et1 = x - qt;

    step(k) = d;
    sqnrR(k) = 10*log10(mean(x.^2)/mean(er1.^2));
    sqnrT(k) = 10*log10(mean(x.^2)/mean(et1.^2));

    maxER(k) = max(abs(er1));
    maxET(k) = max(abs(et1));
end

%% 4. DISPLAY RESULTS
fprintf('\n========== QUANTIZATION RESULTS ==========\n');
```

```
========== QUANTIZATION RESULTS ==========
```

```
fprintf('Amplitude = %.1f\n',A);
```

```
Amplitude = 1.0
```

```
fprintf('Frequency = %d Hz\n',f);
```

```
Frequency = 50 Hz
```

```
fprintf('Sampling Frequency = %d Hz\n',Fs);
```

```
Sampling Frequency = 1000 Hz
```

```
fprintf('Number of Samples = %d\n',N);
```

```
Number of Samples = 1000
```

```
fprintf('Word Length = %d bits\n',B);
```

```
Word Length = 8 bits
```

```
fprintf('8-bit Step Size = %.7f\n',D);
```

```
8-bit Step Size = 0.0078125
```

```

fprintf('\nRounding SQNR = %.4f dB\n',SQr);
```

```
Rounding SQNR = 48.9339 dB
```

```
fprintf('Truncation SQNR = %.4f dB\n',SQt);
```

```
Truncation SQNR = 45.2931 dB
```

```
fprintf('Output SQNR = %.4f dB\n',SQout);
```

```
Output SQNR = 48.9339 dB
```

```

fprintf('Maximum Rounding Error = %.6f\n',max(abs(er)));
```

```
Maximum Rounding Error = 0.003483
```

```
fprintf('Maximum Truncation Error = %.6f\n',max(abs(et)));
```

```
Maximum Truncation Error = 0.005744
```

```
%% 5. FIRST 10 SAMPLES TABLE
T1 = table(n(1:10)',x(1:10)',xr(1:10)', ...
    er(1:10)',xt(1:10)',et(1:10)', ...
    'VariableNames',{'n','Original','Rounded', ...
    'RoundError','Truncated','TruncError'});

disp('First 10 Samples:');
```

```
First 10 Samples:
```

```
disp(T1);
```

n	Original	Rounded	RoundError	Truncated	TruncError
0	0	0	0	0	0
1	0.30902	0.3125	-0.003483	0.30469	0.0043295
2	0.58779	0.58594	0.0018478	0.58594	0.0018478
3	0.80902	0.8125	-0.003483	0.80469	0.0043295
4	0.95106	0.95312	-0.0020685	0.94531	0.005744
5	1	1	0	1	0
6	0.95106	0.95312	-0.0020685	0.94531	0.005744
7	0.80902	0.8125	-0.003483	0.80469	0.0043295
8	0.58779	0.58594	0.0018478	0.58594	0.0018478
9	0.30902	0.3125	-0.003483	0.30469	0.0043295

```
%% 6. WORD-LENGTH TABLE
T2 = table(bits',step',sqnrR',sqnrT',maxER',maxET', ...
    'VariableNames',{'Bits','StepSize','SQNR_Round_dB', ...
    'SQNR_Trunc_dB','MaxError_Round','MaxError_Trunc'});

disp('Word-Length Comparison:');
```

```
Word-Length Comparison:
```

```
disp(T2);
```

Bits	StepSize	SQNR_Round_dB	SQNR_Trunc_dB	MaxError_Round	MaxError_Trunc
4	0.125	23.667	20.871	0.059017	0.087785
8	0.0078125	48.934	45.293	0.003483	0.005744
12	0.00048828	78.811	65.891	0.00011536	0.00042324
16	3.0518e-05	99.697	91.599	1.382e-05	2.6516e-05

```
%% 7. OUTPUT PARAMETER TABLE
Parameter = {
    'Quantized signal'
```

```
    'Quantization error'
    'SQNR'
    'Word-length comparison'
};

Observation = {
    sprintf('8-bit rounding, Delta = %.7f',D)
    sprintf('Maximum rounding error = %.6f',max(abs(er)))
    sprintf('Rounding = %.4f dB; Truncation = %.4f dB',SQr,SQt)
    'SQNR generally increases with word length'
};

Unit = {
    'Amplitude'
    'Amplitude'
    'dB'
    'Bits / dB'
};

T3 = table(Parameter,Observation,Unit);

disp('Output Parameter Table:');
```

```
Output Parameter Table:
```

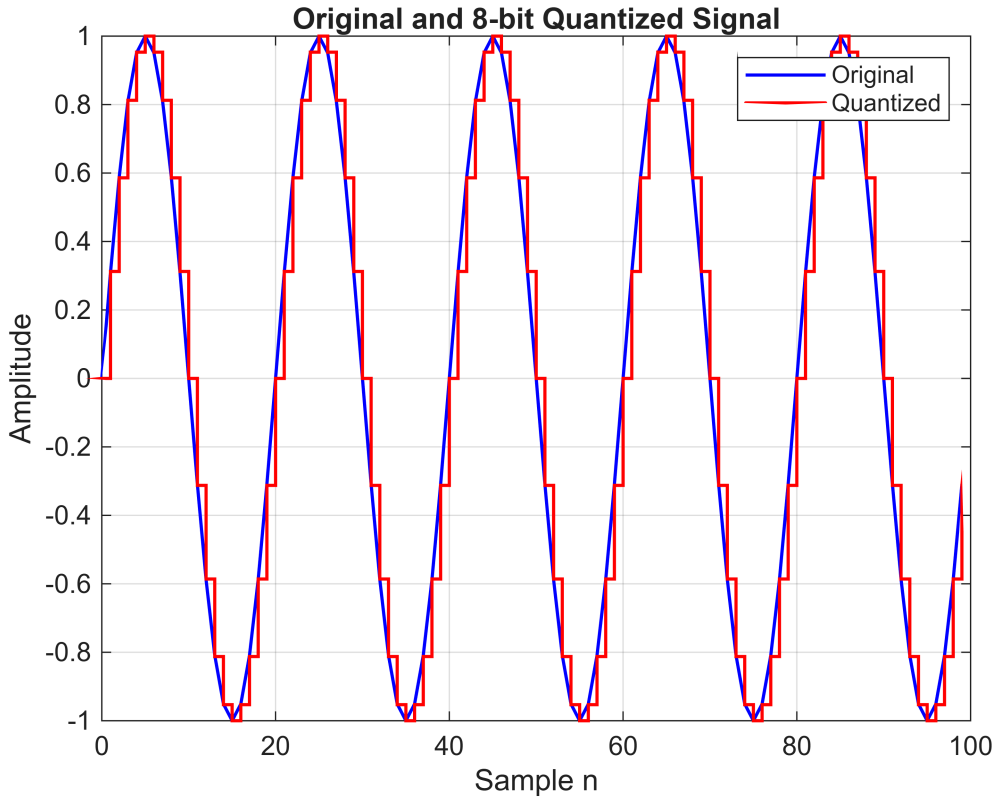
```
disp(T3);
```

Parameter	Observation	Unit
{'Quantized signal'}	{'8-bit rounding, Delta = 0.0078125'}	{'Amplitude'}
{'Quantization error'}	{'Maximum rounding error = 0.003483'}	{'Amplitude'}
{'SQNR'}	{'Rounding = 48.9339 dB; Truncation = 45.2931 dB'}	{'dB'}
{'Word-length comparison'}	{'SQNR generally increases with word length'}	{'Bits / dB'}

```
%% 8. GRAPH 1: ORIGINAL AND QUANTIZED SIGNAL
figure('Name','Original and Quantized Signal');

plot(n(1:100),x(1:100),'b','LineWidth',1.2);
hold on;
stairs(n(1:100),xr(1:100),'r','LineWidth',1.2);

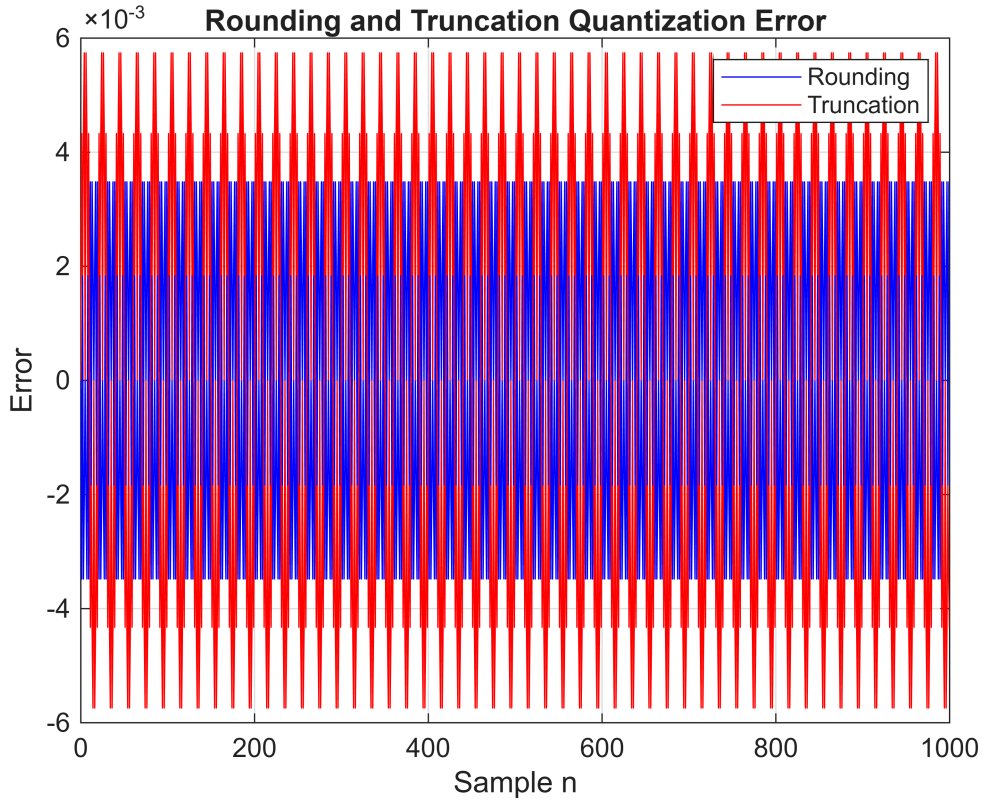
grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Original and 8-bit Quantized Signal');
legend('Original','Quantized');
```




```
%% 9. GRAPH 2: QUANTIZATION ERROR
figure('Name','Quantization Error');

plot(n,er,'b');
hold on;
plot(n,et,'r');

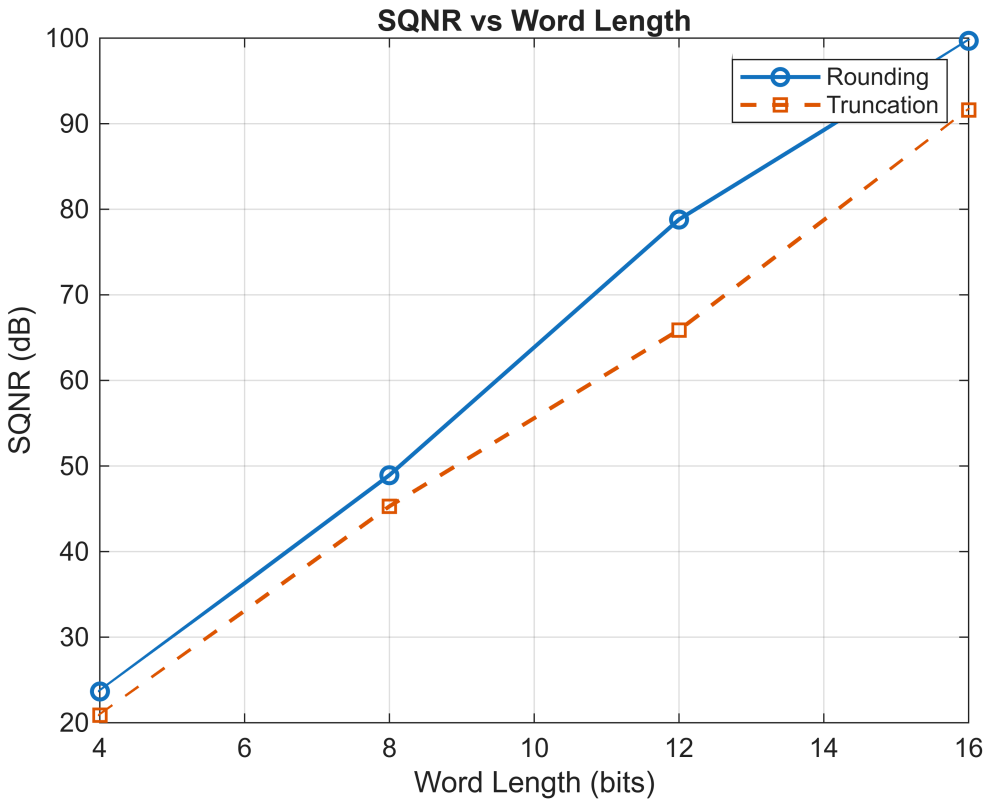
grid on;
xlabel('Sample n');
ylabel('Error');
title('Rounding and Truncation Quantization Error');
legend('Rounding','Truncation');
```




```
%% 10. GRAPH 3: SQNR VS WORD LENGTH
figure('Name','SQNR vs Word Length');

plot(bits,sqnrR,'o-','LineWidth',1.5);
hold on;
plot(bits,sqnrT,'s--','LineWidth',1.5);

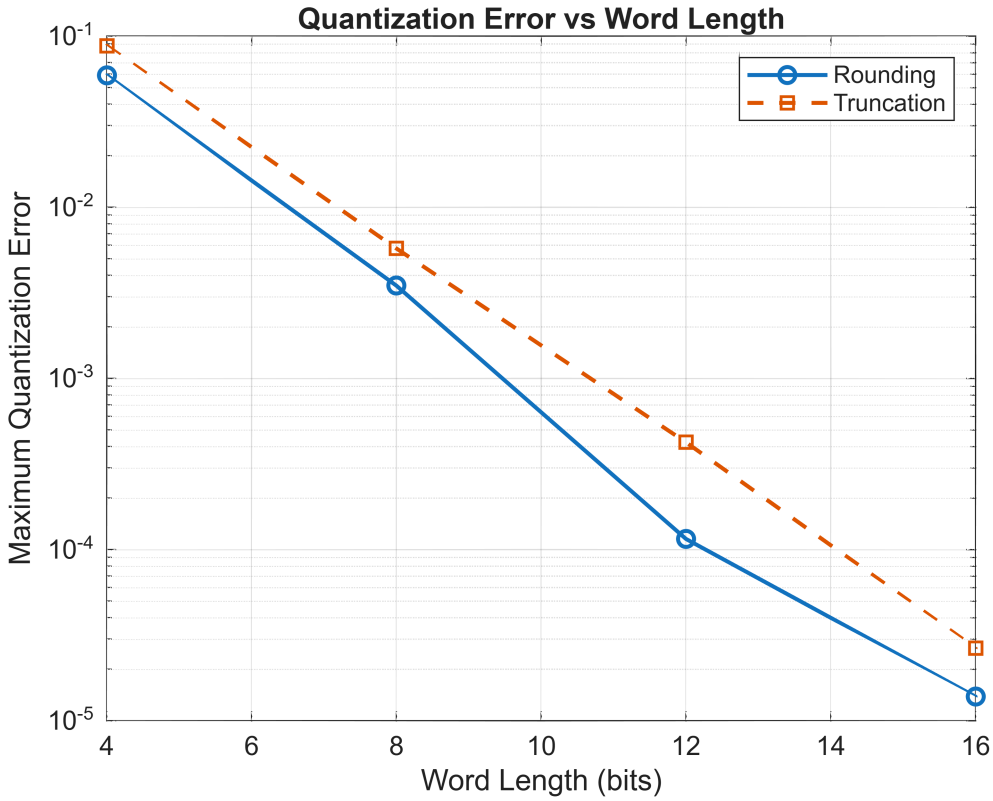
grid on;
xlabel('Word Length (bits)');
ylabel('SQNR (dB)');
title('SQNR vs Word Length');
legend('Rounding','Truncation');
```




```
%% 11. GRAPH 4: MAXIMUM ERROR VS WORD LENGTH
figure('Name','Maximum Quantization Error');

semilogy(bits,maxER,'o-','LineWidth',1.5);
hold on;
semilogy(bits,maxET,'s--','LineWidth',1.5);

grid on;
xlabel('Word Length (bits)');
ylabel('Maximum Quantization Error');
title('Quantization Error vs Word Length');
legend('Rounding','Truncation');
```




```
%% 12. FINAL RESULT
fprintf('\n========== FINAL RESULT ==========\n');
```

```
========== FINAL RESULT ==========
```

```
fprintf('1. Signal quantized using rounding and truncation.\n');
```

```
1. Signal quantized using rounding and truncation.
```

```
fprintf('2. SQNR calculated for both methods.\n');
```

```
2. SQNR calculated for both methods.
```

```
fprintf('3. Word lengths compared: 4, 8, 12 and 16 bits.\n');
```

```
3. Word lengths compared: 4, 8, 12 and 16 bits.
```

```
fprintf('4. Quantization error and SQNR graphs generated.\n');
```

```
4. Quantization error and SQNR graphs generated.
```

```

```